

Souhail Mbarki

AI Engineer | Software Engineering Student

 souhayelmbarki15@gmail.com  +216 52 864 293  Tunis, Tunisia  LinkedIn  Portfolio  Github

 Codeforces  LeetCode  Kaggle

Profile

Software Engineering student specializing in AI Engineering, with hands-on experience building LLM, RAG, agentic AI, and machine learning systems. Experienced in Python, TensorFlow, FastAPI, and applied ML, with a strong competitive programming background. Seeking an AI/ML Engineering internship to build production-oriented AI systems.

Professional Experience

- AI Engineering Intern, TechNet World** 07/2026 – 08/2026
- Developed a modular software system for transforming natural-language business rules into structured Boolean expressions.
 - Designed backend pipelines for retrieval, structured processing, validation, and automated testing.
 - Integrated **Z3-based formal validation** to detect conflicting, ambiguous, and invalid outputs.
 - Debugged complex failure cases and improved system reliability through systematic testing and evaluation.

Projects

- Agentic LLMOps Platform, In Progress**
- Developing a modular software platform for running, evaluating, and monitoring AI applications using Python, FastAPI, MLflow, and Docker.
 - Building backend components for experiment tracking, performance measurement, automated evaluation, and system observability.
 - Designing reusable testing and observability components to improve software reliability and maintainability.
 - Implemented GitHub Actions CI/CD workflows for automated testing and validation.
- AI-Powered Career Assistant Platform, AuraHire**
- Built a full-stack AI platform using **FastAPI, RAG, NLP, and vector search** for context-aware career and interview assistance.
 - Designed REST APIs, retrieval pipelines, and backend services for processing unstructured data and LLM-generated responses.
 - Integrated authentication and containerized the application with **Docker** for reproducible deployment.
- Codeforces Performance Analytics Platform**
- Built a data-processing pipeline to collect, clean, and engineer features from users, contests, ratings, and submissions.
 - Developed analytical workflows to process competitive-programming performance data and identify performance patterns.
 - Trained and evaluated Random Forest, XGBoost, and SVM models and generated personalized recommendations from model predictions.

Skills

- Programming:** C++, Python, SQL
- C++:** STL, OOP, Templates, Algorithms, Data Structures, Memory Management
- Algorithms & Software Engineering:** Data Structures, Algorithms, Competitive Programming, Object-Oriented Programming, Problem Solving
- Backend & Systems:** FastAPI, REST APIs, Docker, Git, GitHub, GitHub Actions
- AI/ML:** PyTorch, TensorFlow, scikit-learn, LLMs, RAG, NLP, Agno
- Data:** Pandas, NumPy, Data Processing, Feature Engineering
- MLOps & Tooling:** MLflow, Model Evaluation, Experiment Tracking

Competitive Programming & Awards

- 2× Tunisian Collegiate Programming Contest (TCPC) Finalist — 13th place** 
- ODC Champions 2025 — Orange Digital Center (5th in Tunisia, 19th in Africa)**

Education

- Software Engineering, National Engineering School of Tunis (ENIT)** 2024 – Expected 2027
- Preparatory classes in Mathematics and Physics, IPEIM** 2022 – 2024

Leadership & Certifications

IEEE ENIT SIGHT Group - Treasurer, 2025-2026 | Efficient LLM Customization  | CCNA: Introduction to Networks 

Languages

- Arabic : Native
- English : B2
- French : B2